

typcas v2 Unified Demo

Simplify

$$\frac{x+1}{x+1} + \ln(e^x) + \sin(x)^2 + \cos(x)^2 + \sqrt{x^2} + \frac{2}{5} + \frac{7}{15} - \frac{13}{15} + (x+4) \cdot (x-4) - x^2 - 16 - 3 + |x| \Rightarrow x - 1$$

Differentiate

$$1$$

Integrate derivative

$$x + C$$

Solve simplified = 0

$$r_1 = 1$$

Trace simplify

$$\begin{aligned} & \frac{x+1}{x+1} + \ln(e^x) + \sin(x)^2 + \cos(x)^2 + \sqrt{x^2} + \frac{2}{5} + \frac{7}{15} - \frac{13}{15} + (x+4) \cdot (x-4) - x^2 - 16 - 3 + |x| \\ &= \frac{x+1}{x+1} + \ln(e^x) + \sin(x)^2 + \cos(x)^2 + x + \frac{2}{5} + \frac{7}{15} - \frac{13}{15} + (x+4) \cdot (x-4) - x^2 - 16 - 3 + x \Rightarrow (x-4) \cdot (x+4) + 15 - x^2 + x \quad (\text{Complexity reduced}) \end{aligned}$$

Restrictions carried from original expression:

$$e^x > 0$$

$\ln(u)$ requires $u > 0$.

Trace integrate (u-sub demo)

$$\int 2x \cdot \cos(x^2) dx$$

$$\text{set } u_1 = x^2$$

$$\text{derive } du_1 = x \cdot 2$$

$$= \int 2x \cdot \cos(x^2) dx \Rightarrow \int \cos(u_1) du_1 \quad (\text{Apply u-substitution})$$

$$= \int \cos(u_1) du_1 \mapsto \sin(u_1) + C \quad (\text{Integrate in } u_1)$$

$$= \sin(u_1) + C \Rightarrow \sin(x^2) + C \quad (\text{Back-substitute})$$